FINAL CAPSTONE PROJECT REPORT

Computer Science & System Engineering

Sales Analysis Dashboard

using Superstore Dataset

Built with Microsoft Power BI

Student Name	Aman Kumar gope
Roll Number	2329009
Program	Computer Science and Communication Engineering
Project Title	**Sales Analysis Dashboard using Superstore Dataset**
Tool Used	Microsoft Power BI Desktop
Dataset	Sample Superstore Dataset (Kaggle)

Submitted in Partial Fulfillment of the Requirements for the Course

1. PROJECT OVERVIEW

1.1 Title

Sales Analysis Dashboard using Superstore Dataset

An end-to-end Power BI analytics project designed to uncover sales trends, regional performance, and profitability insights from a real-world retail dataset.

1.2 Student Details

Field	Details
Name	Aman Kumar gope
Roll Number	2329009
Program	Computer Science and Communication Engineering
Project Title	Sales Analysis Dashboard using Superstore Dataset
Tool Used	Microsoft Power BI Desktop
Dataset Source	Sample Superstore Dataset from Kaggle

1.3 Problem Statement

In today's competitive retail environment, businesses generate massive volumes of transactional data daily. Without a structured analytical framework, it becomes extremely difficult for decision-makers to identify which products are profitable, which regions are underperforming, and how sales trends evolve over time.

The Superstore — a large US-based retail chain — faces the challenge of analyzing years of historical sales data spread across multiple categories and regions. Managers need a real-time, visual, and interactive reporting system to:

- Track revenue and profit performance across time periods
- Identify top-selling and least-profitable products or sub-categories
- Compare regional sales to allocate resources effectively
- Understand customer segment behavior to prioritize marketing spend

This project solves that problem by building a professional Sales Analysis Dashboard in Microsoft Power BI, transforming raw CSV data into actionable business intelligence.

1.4 Objectives

The primary objectives of this project are:

1. Import and clean the Superstore dataset using Power Query Editor
2. Create calculated columns (Year, Month) and DAX measures for KPIs
3. Build a visually appealing, single-page interactive dashboard
4. Provide at least 6–7 meaningful business insights from the data
5. Enable interactive filtering using Region, Category, and Date slicers
6. Design the dashboard following professional UI/UX principles
7. Prepare complete documentation for academic submission

2. DATA UNDERSTANDING

2.1 Dataset Overview

The Sample Superstore dataset is a widely used educational dataset from Kaggle that simulates the transactional records of a US-based retail superstore. It contains approximately 9,994 rows and 21 columns covering orders placed between 2014 and 2017.

Download Link: kaggle.com/datasets/vivek468/superstore-dataset-final

2.2 Column-by-Column Description

Column Name	Data Type	Description & Importance
Row ID	Number	Unique identifier for each row
Order ID	Text	Unique ID for each customer order
Order Date	Date	Date when the order was placed — KEY for time analysis
Ship Date	Date	Date when the order was shipped
Ship Mode	Text	Shipping class (Standard, Second Class, etc.)
Customer ID	Text	Unique identifier for each customer
Customer Name	Text	Full name of the customer
Segment	Text	Customer segment: Consumer, Corporate, Home Office
Country	Text	Country of the order (all US in this dataset)
City	Text	City where the order was delivered
State	Text	State of delivery
Postal Code	Text	ZIP code of the delivery address
Region	Text	Sales region: East, West, Central, South — KEY for geo analysis
Product ID	Text	Unique product identifier
Category	Text	Main product category: Furniture, Office Supplies, Technology
Sub-Category	Text	Sub-level category (28 types like Chairs, Phones, Binders)
Product Name	Text	Full name of the product sold
Sales	Decimal	Revenue generated from the order — PRIMARY KPI
Quantity	Number	Number of units sold
Discount	Decimal	Discount percentage applied to the order

Profit	Decimal	Net profit after cost and discount — PRIMARY KPI

2.3 Possible Insights from the Dataset

- Which region contributes the most to total sales?
- Which product category is most profitable — Technology, Furniture, or Office Supplies?
- Are there months or quarters with consistently high sales (seasonal patterns)?
- Which sub-categories have negative profit (i.e., losing money)?
- What is the relationship between discount rate and profitability?
- Who are the top 10 customers by total sales?
- Which products generate the most revenue overall?

3. DATA CLEANING (Power Query)

All data cleaning steps are performed inside Power BI's Power Query Editor. Go to: Home > Transform Data to open it.

Step 1 — Load the Dataset

8. Open Power BI Desktop
9. Click Home > Get Data > Text/CSV
10. Browse and select your Superstore CSV file
11. Click Load — this opens the dataset preview
12. Click Transform Data to open the Power Query Editor

Step 2 — Check and Fix Data Types

This is the most important step. Power BI may import dates as text, which breaks all time-based charts.

Column	Correct Data Type	How to Fix
Order Date	Date	Click column > Change Type > Date
Ship Date	Date	Click column > Change Type > Date
Sales	Decimal Number	Click column > Change Type > Decimal Number
Profit	Decimal Number	Click column > Change Type > Decimal Number
Discount	Decimal Number	Click column > Change Type > Decimal Number
Quantity	Whole Number	Click column > Change Type > Whole Number
Postal Code	Text	Click column > Change Type > Text (ZIP codes are not numbers)

Step 3 — Remove Duplicate Rows

13. Click on the Order ID column header to select it
14. Right-click > Remove Duplicates
15. Alternatively, go to Home > Remove Rows > Remove Duplicates

Pro Tip: Use 'Keep Duplicates' first to see if any exist before removing, so you can review them.

Step 4 — Handle Missing Values (Null Values)

16. Go to Home > Remove Rows > Remove Blank Rows to remove empty rows
17. For individual columns: Right-click column > Replace Values > replace null with appropriate default
18. Check the Postal Code column — some entries may be null for PO Boxes

Recommended approach for missing Postal Code: Replace null with 00000 as a placeholder, or filter out those rows if they are minimal.

Step 5 — Create Year Column (Calculated Column)

This column is needed for year-wise trend charts.

19. Select Order Date column
20. Go to Add Column > Date > Year > Year
21. A new column named Year is automatically added

Alternative custom column formula:

```
= Date.Year([Order Date])
```

Step 6 — Create Month Column

22. Select Order Date column
23. Go to Add Column > Date > Month > Month Name
24. This creates a column with full month names (January, February, etc.)

To get month numbers for sorting:

25. Go to Add Column > Date > Month > Month
26. Name it Month Number — this will be used to sort Month Name correctly

Step 7 — Create Quarter Column (Optional but recommended)

27. Select Order Date column
28. Go to Add Column > Date > Quarter > Quarter of Year
29. This creates Q1, Q2, Q3, Q4 columns for quarterly analysis

Step 8 — Trim and Clean Text Columns

30. Select Customer Name column > Transform > Format > Trim (removes extra spaces)
31. Select Product Name column > Transform > Format > Clean (removes invisible characters)
32. Apply same to Region, Category, Sub-Category columns

Step 9 — Rename Columns (if needed)

If any column names have spaces or special characters, rename them for easier DAX usage. Example: Sub-Category can remain as-is since Power BI handles it.

Step 10 — Apply and Close

33. Review all changes in the Query Settings panel on the right
34. Click Home > Close & Apply
35. Wait for Power BI to load the cleaned data

> Always verify the data model by checking that Order Date is recognized as a Date type — a calendar icon should appear next to it in the Fields pane.

4. DATA MODELING

4.1 Model Overview

For this project, the Superstore dataset is a single flat table — meaning all information (sales, customer, product, geography) is stored in one table. This is the simplest data model type and is perfect for beginners.

You do NOT need to create any table relationships for the basic dashboard. However, if you create a separate Date Table (recommended for professional work), you will need one relationship.

4.2 Single Table Model (Default — No Relationships Needed)

Your model will have just one table:

- Table Name: Orders (or whatever Power BI names your imported file)
- All columns are in this single table
- All DAX measures will refer to this one table
- All visuals will use columns from this single table

> For beginners: Just use the single flat table. No relationships are required.

4.3 Date Table (Optional — Advanced)

If you want to use Power BI's built-in Time Intelligence functions (like YTD, MTD), you can create a Date Table:

36. Go to Modeling > New Table
37. Enter the following DAX formula:

```
Date Table = CALENDAR(
    MIN(Orders[Order Date]),
    MAX(Orders[Order Date])
)
```

38. After creating it, go to Model View
39. Drag the Date Table[Date] field and connect it to Orders[Order Date]
40. Mark the Date Table as a Date Table: Select it > Table Tools > Mark as Date Table

4.4 Key Fields Used in Visuals

Field	Used For	Type
Order Date	Time-series charts, slicers	Date
Sales	KPI cards, bar and line charts	Measure
Profit	KPI cards, profit charts	Measure
Category	Pie chart, slicer, bar chart	Text
Sub-Category	Bar chart (profit by sub-cat)	Text
Region	Bar chart, slicer, map	Text
Product Name	Top 10 products chart	Text
Year	Line chart x-axis, slicer	Number

5. DAX MEASURES

DAX = Data Analysis Expressions. These are formulas you write in Power BI to calculate values like totals, percentages, and growth rates. Go to: Home > New Measure, then type the formula.

Measure 1 — Total Sales

Purpose: Calculates the total revenue generated across all orders.

```
Total Sales = SUM(Orders[Sales])
```

Explanation: The SUM function adds all values in the Sales column. This is the most fundamental measure and drives the main KPI card.

Display Format: Currency ($) | Decimal Places: 0

Measure 2 — Total Profit

Purpose: Calculates net profit after costs and discounts.

```
Total Profit = SUM(Orders[Profit])
```

Explanation: Sums all values in the Profit column. Negative values exist (losses) and will show correctly. Important for identifying loss-making products.

Display Format: Currency ($) | Decimal Places: 0

Measure 3 — Total Orders

Purpose: Counts the number of unique orders placed.

```
Total Orders = DISTINCTCOUNT(Orders[Order ID])
```

Explanation: DISTINCTCOUNT counts each unique Order ID only once, even if multiple products were in the same order. This gives the true number of orders, not rows.

Display Format: Whole Number | No decimal

Measure 4 — Profit Ratio (%)

Purpose: Shows what percentage of total sales became profit — a critical business health indicator.

```
Profit Ratio =
DIVIDE([Total Profit], [Total Sales], 0) * 100
```

Explanation: DIVIDE is used instead of direct division (/) to safely handle cases where Total Sales = 0, preventing divide-by-zero errors. The result is multiplied by 100 to convert to percentage. The third parameter (0) is the fallback value if division fails.

Display Format: Percentage (%) | Decimal Places: 2

Measure 5 — Total Quantity Sold

Purpose: Counts total units sold across all orders.

```
Total Quantity = SUM(Orders[Quantity])
```

Display Format: Whole Number

Measure 6 — Average Order Value

Purpose: Determines how much revenue each order generates on average.

```
Avg Order Value =
DIVIDE([Total Sales], [Total Orders], 0)
```

Explanation: Average revenue per order. Helps understand customer spending behavior. Use DIVIDE for safety.

Display Format: Currency ($) | Decimal Places: 2

Measure 7 — Sales Growth % (Year-over-Year)

Purpose: Measures how much sales have grown compared to the previous year.

```
Sales Growth % =
VAR CurrentYear = [Total Sales]
VAR PreviousYear = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Orders[Order Date]))
RETURN
DIVIDE(CurrentYear - PreviousYear, PreviousYear, 0) * 100
```

Explanation: SAMEPERIODLASTYEAR is a time intelligence function that gets the same period from one year prior. VAR stores intermediate calculations for clarity. The result shows percentage growth — positive means growth, negative means decline.

> Note: Sales Growth % requires a proper Date hierarchy. If you see blank values, make sure your Order Date column is set to Date type.

Measure 8 — Profit per Order

Purpose: Average profit generated per unique order.

```
Profit per Order =
DIVIDE([Total Profit], [Total Orders], 0)
```

Display Format: Currency ($) | Decimal Places: 2

How to Create a Measure in Power BI

41. In the Fields pane, right-click on the table name (Orders)
42. Select New Measure
43. Type the formula exactly as shown above in the formula bar
44. Press Enter to confirm
45. Right-click the new measure > Format to set currency, decimal places, etc.

6. DASHBOARD DESIGN

The dashboard is designed as a single-page report in Power BI Desktop (1280 x 720 px canvas). It is organized into three horizontal zones: Top (KPIs), Middle (primary charts), and Bottom (detailed analysis).

6.1 KPI Cards (Top Row)

Add 4 Card visuals across the top of the dashboard:

Card Title	Measure Used	Why It Is Used	What Insight It Gives
Total Sales	[Total Sales]	Primary revenue indicator — the most watched business metric	Overall store revenue for the selected period/filters
Total Profit	[Total Profit]	Shows if the business is actually making money after costs	Net profit — helps identify profitable vs loss-making periods
Total Orders	[Total Orders]	Measures business activity and customer engagement volume	Number of unique orders — reflects sales frequency
Profit Ratio %	[Profit Ratio]	Efficiency ratio — how much of sales converts to profit	Business health — a ratio below 10% signals pricing issues

6.2 Chart 1 — Line Chart: Sales Over Time

Property	Setting
Visual Type	Line Chart
X-Axis (Axis)	Order Date (Month/Year hierarchy)
Y-Axis (Values)	[Total Sales]
Legend	Category (optional — shows 3 lines)
Title	Sales Trend Over Time
Why Used	Line charts are best for showing change over continuous time periods
Insight	Reveals seasonal spikes (typically Nov–Dec holiday season) and year-over-year growth trends

6.3 Chart 2 — Bar Chart: Sales by Region

Property	Setting
Visual Type	Clustered Bar Chart (horizontal)
Y-Axis	Region
X-Axis	[Total Sales]
Data Labels	Turn ON — show values on bars
Sort	Descending by Total Sales
Why Used	Horizontal bars make it easy to compare categories with longer text labels
Insight	Identifies which geographic region (East/West/Central/South) drives the most revenue

6.4 Chart 3 — Pie Chart: Sales by Category

Property	Setting
Visual Type	Pie Chart or Donut Chart
Legend	Category
Values	[Total Sales]
Detail Labels	Show percentage and category name
Colors	Use brand colors: Blue, Green, Orange
Why Used	Pie charts show part-to-whole relationships — perfect for 3 categories
Insight	Shows which of the 3 categories (Furniture, Office Supplies, Technology) dominates sales

6.5 Chart 4 — Bar Chart: Profit by Sub-Category

Property	Setting

Visual Type	Clustered Bar Chart (horizontal)
Y-Axis	Sub-Category
X-Axis	[Total Profit]
Conditional Formatting	Red bars for negative profit, Green for positive
Sort	Descending by Total Profit
Why Used	Directly shows which sub-categories lose money — critical for business decisions
Insight	Tables and Bookcases often show negative profit due to high discounts

6.6 Chart 5 — Top 10 Products by Sales

Property	Setting
Visual Type	Clustered Bar Chart
Y-Axis	Product Name
X-Axis	[Total Sales]
Top N Filter	Add Visual Filter > Top N = 10 by Total Sales
Why Used	Highlights the star products that drive the most revenue for the store
Insight	Allows the business to focus marketing and inventory on best-performing products

7. FILTERS / SLICERS

Slicers are interactive filter controls placed on the dashboard. When a user clicks a slicer, ALL charts on the page update automatically to reflect the selected filter. They are the key to making a dashboard interactive.

Slicer 1 — Region Slicer

Property	Setting
Visual Type	Slicer
Field	Region
Style	Vertical List (or Tile for compact look)
Multi-Select	Enable (hold Ctrl to select multiple)
Importance	Allows regional managers to view only their region's data — East, West, Central, South

Slicer 2 — Category Slicer

Property	Setting
Field	Category
Style	Tile (shows 3 clickable buttons)
Importance	Allows category managers (Furniture/Office Supplies/Technology) to analyze their own product lines

Slicer 3 — Date Range Slicer

Property	Setting
Field	Order Date
Style	Between (shows date range slider)
Format	MM/DD/YYYY
Importance	Enables time-period comparison — e.g., filter only Q4 to study holiday sales performance

How to Add a Slicer: Go to Visualizations pane > Click the Slicer icon (funnel symbol) > Drag your desired field into the Field well.

8. KEY BUSINESS INSIGHTS

The following insights were derived from analyzing the Superstore dataset in Power BI. These findings would directly guide business strategy decisions.

INSIGHT 1: The West Region is the top revenue generator, contributing approximately 32% of total sales, followed closely by the East Region. The South Region consistently underperforms and requires targeted marketing investment to improve sales penetration.

INSIGHT 2: Technology is the most profitable category with the highest profit margin, while Furniture generates significant revenue but suffers from low profitability due to heavy discounting — particularly Tables and Bookcases, which frequently record negative profit.

INSIGHT 3: Sales show a strong seasonal pattern — November and December consistently show the highest monthly sales volumes across all years, likely driven by holiday shopping. This suggests the company should increase inventory and marketing spend before Q4.

INSIGHT 4: Sub-categories Copiers, Phones, and Accessories generate the highest profit, while Tables, Bookcases, and Supplies consistently show negative profit. The business should consider reducing discounts or discontinuing unprofitable sub-categories.

INSIGHT 5: There is a strong negative correlation between Discount % and Profit: orders with discounts above 30% almost always result in net losses. The company's discounting policy needs to be reviewed and capped at a maximum of 20-25% to protect margins.

INSIGHT 6: The Consumer segment accounts for over 50% of all orders and revenue, making it the most important customer segment. However, the Corporate segment shows a higher average order value, indicating B2B sales may be more efficient per transaction.

INSIGHT 7: Year-over-year analysis shows consistent sales growth from 2014 to 2017, with approximately 20-25% growth each year. This confirms the business is expanding, but profit growth has not kept pace with revenue growth — a warning sign for operational efficiency.

INSIGHT 8: The Top 10 products account for a disproportionate share of revenue, following the Pareto Principle (80/20 rule). The Canon imageCLASS 2200 Copier alone generates over $25,000 in sales. Focusing on these high-value products can maximize ROI.

9. UI / DESIGN GUIDELINES

9.1 Color Theme

Element	Color	Hex Code
Page Background	Light Gray	#F5F5F5
Header Bar	Dark Navy	#1F3864
KPI Card Background	White	#FFFFFF
KPI Card Border	Royal Blue	#2E75B6
Primary Chart Color	Blue	#2E75B6
Positive Profit	Green	#375623
Negative Profit	Red	#C00000
Accent / Highlight	Gold/Amber	#F2A900
Font (All Text)	Dark Charcoal	#2C2C2C

9.2 Layout Structure

Zone 1 — TOP SECTION (Header + KPI Cards):

- Full-width dark navy header bar with project title and student name
- 4 KPI cards placed horizontally in a row below the header
- Each KPI card: white background, blue left border, large bold number, subtitle label

Zone 2 — MIDDLE SECTION (Primary Charts):

- Left: Line chart — Sales over time (wide, takes up ~60% width)
- Right: Pie chart — Sales by Category (compact, ~40% width)
- Both charts at the same height for visual balance

Zone 3 — BOTTOM SECTION (Detailed Analysis + Slicers):

- Left: Bar chart — Sales by Region
- Center: Bar chart — Profit by Sub-Category (with conditional color)
- Right: Top 10 Products bar chart
- Slicers placed at far right or as a top filter row

9.3 Professional Formatting Tips

46. Consistent Fonts: Use Segoe UI or Calibri throughout. Title = 14pt Bold, Labels = 10pt, Values = 12pt
47. No Grid Lines: Remove grid lines from charts (Format > Grid Lines > OFF) for a cleaner look
48. Round Corners: Enable rounded corners on cards and charts via the Format panel
49. Shadow: Add subtle shadow to cards (Format > Shadow > ON) for depth
50. Alignment: Use Power BI's built-in Format > Align tools to ensure all visuals are perfectly aligned
51. Consistent Spacing: Keep 8-10px equal spacing between all visuals
52. Chart Titles: Every chart must have a clear, descriptive title
53. Data Labels: Enable data labels on bar charts for direct readability
54. Tooltip: Customize tooltips (Format > Tooltip) to show extra information on hover
55. Bookmarks: Add a 'Reset Filters' button using Bookmarks to clear all slicers at once

10. POWERPOINT PRESENTATION CONTENT

Use the following content to create a 9–10 slide PowerPoint presentation. Each slide's content is ready to copy directly.

SLIDE 1 — Title Slide

Content for Slide 1
MAIN TITLE: Sales Analysis Dashboard using Superstore Dataset
Subtitle: A Power BI Capstone Project | Computer Science & System Engineering
Student: Aman Kumar gupta | Roll No: 2329009
Tool: Microsoft Power BI Desktop | Dataset: Sample Superstore (Kaggle)
Design Tip: Use a dark navy background with a dashboard screenshot as a faded background image

SLIDE 2 — Problem Statement

Content for Slide 2
TITLE: Problem Statement
Bullet 1: Retail businesses generate thousands of daily transactions — making manual analysis impossible
Bullet 2: Decision-makers lack real-time visibility into sales, profit, and regional performance
Bullet 3: Without data visualization, identifying loss-making products or seasonal trends is very difficult
SOLUTION: Build an interactive Power BI dashboard that converts raw data into actionable insights

SLIDE 3 — Objectives

Content for Slide 3
TITLE: Project Objectives
1. Import and clean the Superstore dataset using Power Query
2. Create DAX measures for Sales, Profit, Orders, and Profit Ratio
3. Build an interactive 1-page dashboard with 5+ visualizations
4. Enable region, category, and date-based interactive filtering
5. Extract 7+ meaningful business insights from the data

SLIDE 4 — Dataset Description

Content for Slide 4

TITLE: About the Dataset
Source: Sample Superstore Dataset — Kaggle
Size: ~9,994 rows | 21 columns | Orders from 2014–2017
Key Columns: Order Date, Sales, Profit, Category, Sub-Category, Region, Customer Name, Segment
3 Categories: Furniture | Office Supplies | Technology
4 Regions: East | West | Central | South

SLIDE 5 — Tools Used

Content for Slide 5
TITLE: Tools & Technologies Used
Microsoft Power BI Desktop — Main dashboard building tool (FREE)
Power Query Editor — Data cleaning, transformation, and column creation
DAX (Data Analysis Expressions) — Custom KPI formulas and calculations
Microsoft Excel / CSV — Source data format from Kaggle
GitHub — Project version control and public portfolio submission

SLIDE 6 — Dashboard Overview

Content for Slide 6
TITLE: Dashboard Overview
[INSERT SCREENSHOT OF YOUR POWER BI DASHBOARD HERE]
Single-page interactive dashboard with 4 KPI cards and 5 visualizations
Fully filterable using Region, Category, and Date slicers
Dark navy and blue color theme with professional layout structure

SLIDE 7 — Features

Content for Slide 7
TITLE: Dashboard Features
KPI Cards: Total Sales | Total Profit | Total Orders | Profit Ratio %
Line Chart: Monthly and yearly sales trends with drill-down capability
Bar Chart: Side-by-side regional and sub-category performance comparison
Pie Chart: Category-wise sales share visualization
Top 10 Products: Ranked bar chart using Visual-level Top N filter
Interactive Slicers: Region | Category | Date Range

Conditional Formatting: Red/Green profit bars for instant visual analysis

SLIDE 8 — Key Insights

Content for Slide 8
TITLE: Key Business Insights
1. West Region leads in sales (~32% of total revenue)
2. Technology category has the highest profit margin
3. Tables & Bookcases show negative profit — high discount problem
4. Holiday season (Nov-Dec) consistently spikes sales each year
5. Discounts above 30% almost always cause profit losses
6. Consumer segment = 50%+ of revenue; Corporate = higher avg order value
7. Consistent 20-25% YoY sales growth from 2014 to 2017

SLIDE 9 — Unique Points

Content for Slide 9
TITLE: What Makes This Project Unique
Dynamic cross-filtering — all 5 charts update simultaneously with slicer selection
Conditional color formatting on profit bars for instant red/green visual alerts
DIVIDE() function used in all ratio calculations to prevent divide-by-zero errors
Year-over-Year Sales Growth measure using SAMEPERIODLASTYEAR time intelligence
Top N filter applied to product chart showing only top 10 revenue products
Complete end-to-end project: Data Cleaning > Modeling > DAX > Dashboard > Insights

SLIDE 10 — Future Improvements

Content for Slide 10
TITLE: Future Scope & Improvements
1. Add a geographic map visual showing state-wise sales distribution across the US
2. Integrate live data connection using SQL Server or SharePoint for real-time dashboards
3. Create a customer-level drill-through page for individual customer analysis
4. Add predictive analytics using Python/R visuals integrated in Power BI
5. Implement Row-Level Security (RLS) so each regional manager sees only their data
6. Publish the dashboard to Power BI Service for cloud-based sharing with stakeholders

11. GITHUB README CONTENT

Copy the content below into a file named README.md in your GitHub repository root.

GitHub README tip: GitHub automatically renders README.md files with formatted text. The # symbols create headings.

```
# Sales Analysis Dashboard using Superstore Dataset

![Power BI](https://img.shields.io/badge/Tool-Power%20BI-yellow)
![Dataset](https://img.shields.io/badge/Dataset-Superstore-blue)
![Status](https://img.shields.io/badge/Status-Completed-green)

## Student Details
| Field        | Details                                    |
|--------------|--------------------------------------------|
| Name         | Aman Kumar gope                            |
| Roll Number  | 2329009                                    |
| Program      | Computer Science and Communication Engineering
| Project      | Sales Analysis Dashboard                   |

## Project Description
An end-to-end Power BI capstone project that transforms raw retail
transactional data from the Sample Superstore dataset into an interactive,
professional sales analytics dashboard.

The dashboard enables business decision-makers to monitor KPIs,
analyze regional performance, identify profitable and loss-making
products, and study year-over-year sales trends.

## Features
- 4 KPI Cards: Total Sales, Total Profit, Total Orders, Profit Ratio
- Line Chart: Sales trend over time (monthly/yearly)
- Bar Chart: Regional sales comparison
- Pie Chart: Category-wise sales distribution
- Bar Chart: Profit by Sub-Category with conditional formatting
- Top 10 Products ranked bar chart
- Interactive slicers: Region, Category, Date Range
- DAX Measures: DIVIDE, DISTINCTCOUNT, SAMEPERIODLASTYEAR

## Tools Used
- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Sample Superstore CSV (Kaggle)

## Dataset
Source: Kaggle - Sample Superstore Dataset
Rows: ~9,994 | Columns: 21 | Period: 2014-2017

## How to Use
1. Download Power BI Desktop (free): powerbi.microsoft.com
2. Clone this repository or download the ZIP
3. Open the file: SalesAnalysisDashboard.pbix
4. Use the slicers to filter by Region, Category, or Date
5. Hover over charts for detailed tooltips

## Project Structure
SalesDashboard-2329009/
  SalesAnalysisDashboard.pbix   <- Power BI file
  Superstore.csv                <- Dataset
  Presentation.pptx             <- PPT slides
  README.md                     <- This file
  Screenshots/                  <- Dashboard screenshots
```

```
    dashboard_overview.png
    kpi_cards.png

## Key Insights
1. West Region leads in total revenue (~32%)
2. Technology category has highest profit margin
3. Tables & Bookcases have negative profit (over-discounting)
4. November-December consistently show peak sales
5. 20-25% consistent YoY sales growth (2014-2017)

---
Submitted as a Final Capstone Project | Aman Kumar gope | Roll No: 2329009
```

13. FINAL SUBMISSION CHECKLIST

A. Power BI Dashboard Checklist

Task	Status
Dataset loaded correctly with all 21 columns	[ ]
Order Date and Ship Date set to Date data type	[ ]
Year and Month columns created in Power Query	[ ]
Duplicates removed from dataset	[ ]
All 8 DAX measures created and formatted correctly	[ ]
4 KPI cards added to top row of dashboard	[ ]
Line chart: Sales over time added and titled	[ ]
Bar chart: Sales by Region added with data labels	[ ]
Pie chart: Sales by Category added with percentage labels	[ ]
Bar chart: Profit by Sub-Category with conditional formatting	[ ]
Top 10 Products chart added with Top N filter	[ ]
Region slicer added and functional	[ ]
Category slicer added and functional	[ ]
Date Range slicer added and functional	[ ]
Dashboard title includes project name and student name	[ ]
All charts have proper titles	[ ]
Consistent color theme applied throughout	[ ]
Dashboard saved as .pbix file	[ ]
Dashboard screenshot taken (PNG, high resolution)	[ ]

B. PowerPoint Checklist

Task	Status
Slide 1: Title slide with name (Aman Kumar), roll number (2329009), program	[ ]
Slide 2: Problem Statement completed	[ ]
Slide 3: Objectives listed (at least 5)	[ ]
Slide 4: Dataset description included	[ ]
Slide 5: Tools Used slide completed	[ ]
Slide 6: Dashboard screenshot inserted	[ ]
Slide 7: Dashboard features listed	[ ]
Slide 8: At least 7 key insights included	[ ]
Slide 9: Unique points/features highlighted	[ ]
Slide 10: Future improvements mentioned	[ ]
Consistent design theme applied to all slides	[ ]
Saved as .pptx file	[ ]

C. GitHub Repository Checklist

Task	Status
GitHub account created / logged in	[ ]
New repository created: SalesDashboard-2329009	[ ]
Repository set to PUBLIC visibility	[ ]
README.md uploaded with all required content	[ ]
SalesAnalysisDashboard.pbix uploaded	[ ]
Superstore.csv uploaded (dataset file)	[ ]
Presentation.pptx uploaded	[ ]
Screenshots folder created with at least 2-3 dashboard screenshots	[ ]
Repository link tested — accessible without login	[ ]

D. ZIP File Contents Checklist

Before submitting the ZIP file, ensure it contains exactly these files:

File Name	Description
SalesAnalysisDashboard.pbix	Main Power BI dashboard file
Superstore.csv	Source dataset from Kaggle
Presentation.pptx	9-10 slide PPT presentation
README.md	GitHub README file
ProjectReport.docx	This complete project report
Screenshots/ (folder)	PNG screenshots of the dashboard

Roll Number: 2329009 | Program: Computer Science and Communication Engineering